

Mid-Evaluation Report

Does Repeated Drought Reduce Ecosystem Recovery?

Team Earth 404

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1 Problem Statement

1.1 What is the problem?

Drought is an environmentally significant disturbance capable of altering ecosystem productivity, vegetation development, and an ecosystem's capacity to withstand future climatic stress. Critically, drought's effects do not necessarily end when the rains return: lingering physiological, hydrological, and ecological changes – termed drought legacies – can persist for months or years after a drought event ends. These legacies have the potential to shape how the same ecosystem responds to the next drought it faces.

1.2 Why is this important?

As global climate patterns shift, droughts are projected to become more frequent, more intense, and longer in duration. This raises a practically urgent possibility: an ecosystem may be struck by a new drought before it has had sufficient time to recover from a previous one. If repeated drought exposure compounds damage over time, this has direct implications for:

- **Agricultural planning** in drought-prone regions such as Anantapur and Kurnool, where rain-fed agriculture and groundwater-dependent livelihoods are directly exposed to ecosystem productivity losses.
- **Climate adaptation policy**, since understanding whether repeated drought erodes long-term ecosystem resilience changes how water resources, afforestation, and drought-relief programs should be prioritized.
- **Scientific understanding of ecological memory** – whether ecosystems “remember” past stress in a way that weakens future responses, a question of active interest in global change ecology.

1.3 Our specific research question

Formally, our project investigates:

Does repeated exposure to drought reduce the resistance and recovery capacity of ecosystems?

We decompose this single question into two independently testable sub-questions:

1. **Resistance sub-question:** When a drought strikes an ecosystem that has not yet fully recovered from a prior drought (a “successive” drought), does it cause a greater drop in ecosystem productivity than an isolated drought would?
2. **Recovery sub-question:** Does an ecosystem take longer to return to its normal productivity after a successive drought, compared to after an isolated drought?

We additionally test a third dimension motivated by our literature review: whether this relationship is spatially uniform or varies across ecosystems with different baseline climates – since several studies we reviewed (e.g., Liu et al., 2026) found that drought legacy effects were detectable only in certain regions (dry sub-humid zones) and absent elsewhere. This directly justifies our decision to study six districts spanning a range of climatic and drought conditions rather than one.

2 Data Used

2.1 Study area selection and rationale

We selected six districts in Andhra Pradesh, India, deliberately chosen to span a rainfall and aridity gradient, rather than six similar locations. Three districts were used in our earlier draft; Kadapa, Chittoor, and Prakasam were added for this version to increase the number of drought events and to strengthen the regional comparison.

District	Climatic Profile	Role in Study
Anantapur	Semi-arid interior district; one of the most chronically drought-affected districts in peninsular India	Primary focus district
Kurnool	Semi-arid, moderately dry; adjacent to Anantapur in the same dry belt	Within-region comparison
Kadapa	Semi-arid interior district in the same Rayalaseema dry belt as Anantapur and Kurnool	Within-region comparison (added)
Chittoor	Southern district with a more varied, partly hilly landscape; drier than the coast but less arid than Anantapur	Intermediate case (added)
Prakasam	Coastal-influenced district on the south-eastern coast with a drought-prone climate	Coastal drought-prone contrast (added)
East Godavari	Coastal, tropical-wet, strongly monsoon-influenced	Contrast case to test regional unevenness

Table 1: Study districts and the rationale for each selection

This selection directly operationalizes the “environmentally uneven” dimension of our research question: if drought legacy effects are genuinely dependent on regional climate (as suggested by Liu et al., 2026), a single-district study could not detect this, whereas our six-district design can.

2.2 Datasets extracted

All data were extracted using Google Earth Engine, for each district’s administrative boundary (FAO GAUL 2015, Level 2). Five datasets were extracted per district. All series were aggregated to monthly resolution and restricted to a **common analysis period of January 2001 to December 2022 (264 months per district)**, with no missing values in the final analysis period, so that every district is analysed over exactly the same years.

Dataset	Source Product	Resolution	Purpose
SPEI-03	CSIC SPEIbase v2.10 (monthly SPEI-03 series for each district)	~55 km, monthly	Identifying when and how severely each drought occurred
MODIS GPP	MOD17A2H, V006 (2001–2021) stitched with V061 (2021–2026)	500 m, 8-day	Primary ecosystem-health indicator; used to compute resistance, recovery time, and post-drought responses
ERA5-Land	ECMWF Monthly Aggregates	~11 km, monthly	2 m air temperature; a temperature anomaly is computed for each event but not analysed in this version
MODIS NDVI	MOD13Q1	250 m, 16-day	Independent cross-check of GPP-derived resistance
CHIRPS Rainfall	UCSB-CHG Daily	~5.5 km, daily	Extracted per original scope; not used in the final analysis

Table 2: All datasets extracted, their source, resolution, and role

2.3 How NDVI, rainfall, and soil moisture were used

NDVI and CHIRPS rainfall were collected because they were explicitly listed in our original project brief. Their roles in this version are as follows:

- **Rainfall vs. SPEI:** SPEI (Standardized Precipitation-Evapotranspiration Index) is itself derived from precipitation and temperature data. Including raw CHIRPS rainfall alongside SPEI would introduce redundant, highly correlated information without adding genuinely new explanatory power, so rainfall is not used.
- **NDVI vs. GPP:** Both NDVI (vegetation greenness) and GPP (productivity) measure overlapping aspects of vegetation health. We use GPP as our single primary outcome indicator because it is a more direct, process-based measure of carbon uptake, whereas NDVI is a reflectance-based proxy for greenness. In this version NDVI is used for **independent validation**: we recompute resistance from NDVI and test whether it agrees with GPP-based resistance (Section 3.6).
- **ERA5-Land soil moisture:** soil moisture, which was a candidate predictor in our earlier draft, is not used in this version because we replaced the machine-learning step with direct statistical tests of the repeated-drought effect (Section 3.5). Air temperature is computed per event as a covariate but is not analysed in this version.

3 Methodology

3.1 Step 1: Identifying real drought events

For each district, we examined the monthly SPEI-03 time series and flagged any month where SPEI-03 fell below the threshold of -1.0 as a “drought month” – a standard, widely-used threshold for identifying moderate-to-severe meteorological drought. Consecutive drought months were grouped together into a single drought event, and the following rules were applied:

- **Merging:** drought runs separated by two or fewer non-drought months were merged into one event, since a single-month rise in SPEI does not mean the underlying water deficit has ended.
- **Minimum duration:** runs shorter than two months were discarded.
- **Recorded attributes:** start date (first month below threshold), end date (last month below threshold), duration, minimum SPEI value reached, and *severity*, defined as the sum of $(-SPEI)$ over the drought months.

We deliberately used variable-length events based on actual observed conditions, rather than assuming a fixed duration (e.g., a fixed 3-month window) for every event. Because vegetation responds to water deficit with a delay, the vegetation-response window for each event was extended by one month beyond the drought end date.

3.2 Step 2: Classifying droughts as “successive” or isolated

The definition of “successive” was revised in this version to match the research question more closely. A drought is classified as **successive (repeat)** only if it begins *before the ecosystem has recovered from the previous drought*, and the previous drought actually caused a GPP loss:

- The first drought in a district’s record is always classified as isolated (there is no previous drought).
- If the previous drought caused no GPP loss ($\text{resistance} \geq 0.90$, see Step 3), there is no recovery to wait for, so the new drought is isolated.
- If the previous drought caused a loss and GPP recovered (Step 4) *before* the new drought began, the new drought is isolated.
- If the previous drought caused a loss and the new drought began *before* the recovery date, or the ecosystem never recovered within 24 months, the new drought is classified as successive.

Our earlier draft instead asked whether GPP had climbed back to the district’s overall long-term average during the gap; that criterion was sensitive to ordinary seasonal GPP cycles. The new definition ties “repeat” directly to the measured recovery of the previous event. A fixed 24-month gap definition is retained only as a planned sensitivity analysis. This binary classification is the central explanatory variable for our entire research question: it directly operationalizes “repeated exposure before recovery.”

3.3 Step 3: Quantifying resistance

Monthly GPP was first converted to a ratio of a **seasonal baseline**: the median GPP of the same calendar month, computed from non-drought months only so that drought years do not lower their own reference. For each drought event, we computed a resistance score:

$$\text{Resistance} = \frac{\text{minimum GPP during the drought event (plus 1-month lag)}}{\text{seasonal baseline GPP for that calendar month}} \quad (1)$$

Interpretation: a resistance score near 1.0 means vegetation productivity barely dipped during the drought (highly resistant); a score near 0 means productivity nearly collapsed (highly vulnerable). An event is counted as a genuine **GPP loss** when resistance falls below 0.90. Using a seasonal baseline removes the effect of the normal annual GPP cycle, so an event that begins in a naturally low-GPP month is not mistaken for a severe loss. The same procedure applied to NDVI gives an NDVI-based resistance used for validation.

3.4 Step 4: Quantifying recovery time

For each drought event that caused a GPP loss, we examined the months after the vegetation-response window and counted how many months elapsed until GPP returned to at least 90% of its seasonal baseline for two consecutive months:

$$\text{Recovery Time (months)} = \min\{t : r_t \geq 0.9 \text{ and } r_{t+1} \geq 0.9, t > \text{end of response window}\} \quad (2)$$

where r_t is the GPP-to-baseline ratio in month t . We imposed a maximum observation window of 24 months; any event that had not recovered within this window would have been flagged as “non-recovered” (in practice, all 54 events with a GPP loss recovered within this window – see Results). Events with no GPP loss have no recovery time. Two further metrics describe the post-drought state, based on seasonal GPP z-scores (anomalies relative to the same calendar month):

- **Post-drought residual** (`post_z`): mean GPP z-score in the six months after the response window; values below zero mean GPP remains below normal.
- **Recovery gain**: post-drought residual minus the mean z-score during the drought, i.e. how far GPP rebounds.

3.5 Step 5: Statistical testing of the repeated-drought effect

Our earlier draft used a Random Forest regressor to rank predictors. With the stricter definition of “successive” now yielding very few repeat events (Section 4.1), a flexible machine-learning model has almost nothing to learn about the repeat flag, and its feature importances would be unreliable. We therefore replaced it with direct, transparent tests of repeat versus isolated events for four outcomes (resistance, recovery time, post-drought residual, recovery gain):

1. Mean and median differences with a Mann–Whitney U test and Cliff’s delta effect size.
2. A bootstrap 95% confidence interval for the difference in means (5,000 resamples).
3. A stratified permutation test in which repeat labels are shuffled *within* each district, so that differences in district baselines cannot create a false signal.
4. An OLS regression adjusting for drought severity, duration, and district, with standard errors clustered by start year because droughts in neighbouring districts often belong to the same regional climate episode.

5. Spearman correlations of event order, prior-drought severity, and pre-drought GPP state with each outcome, as a check for cumulative (“legacy”) effects.

Why this change: with so few repeat events, confidence intervals and exact tests communicate the uncertainty honestly, whereas a cross-validated machine-learning score would suggest more precision than the data can support.

3.6 Step 6: Independent cross-verification

To ensure our results were not an artefact of how GPP was measured or when droughts began, we performed two cross-checks:

- **NDVI validation:** Spearman correlation between GPP-based and NDVI-based resistance across all events.
- **Seasonal-artefact check:** a Kruskal–Wallis test of whether resistance differs by the season in which a drought started (winter, pre-monsoon, monsoon, post-monsoon). A large p -value indicates that the seasonal baseline has successfully removed the seasonal cycle from the metric.

4 Results

4.1 Drought events identified

Applying our methodology across all six districts yielded a total of **58 distinct drought events** over 2001–2022:

District	Events	Share of Total	With GPP loss	Repeat events
Anantapur	10	17.2%	10	1
Kurnool	10	17.2%	10	0
East Godavari	11	19.0%	10	1
Kadapa	8	13.8%	8	0
Chittoor	8	13.8%	8	0
Prakasam	11	19.0%	8	0
Total	58	100%	54	2

Table 3: Drought events identified per district, 2001–2022

Two results stand out. First, 54 of the 58 events (93%) produced a GPP loss (resistance below 0.90); the remaining four (one in East Godavari and three in Prakasam) barely registered in vegetation productivity. Second, **all 54 events with a GPP loss recovered to at least 90% of baseline within the 24-month window** – none was classified as non-recovered. Recovery was typically fast: the median recovery time was 2 months, the mean 2.9 months, and 26 of the 54 events (48%) recovered in a single month. The longest recovery was 18 months (Kurnool, drought beginning January 2018). This indicates that, within the 24-month recovery window used in this study, none of the 54 loss-producing drought events remained unrecovered.

Because recovery is often completed within a few months, in many cases the next drought arrives only after recovery has occurred. **Only 2 of the 58 events** were classified as successive: the

third Anantapur event (January 2004, which began before recovery from the 2003 drought in April 2004) and the third East Godavari event (May 2003, which began before recovery from the 2002–03 drought in January 2004). The other 56 events were first or isolated droughts. This imbalance determines how strongly the central hypothesis can be tested, as discussed below.

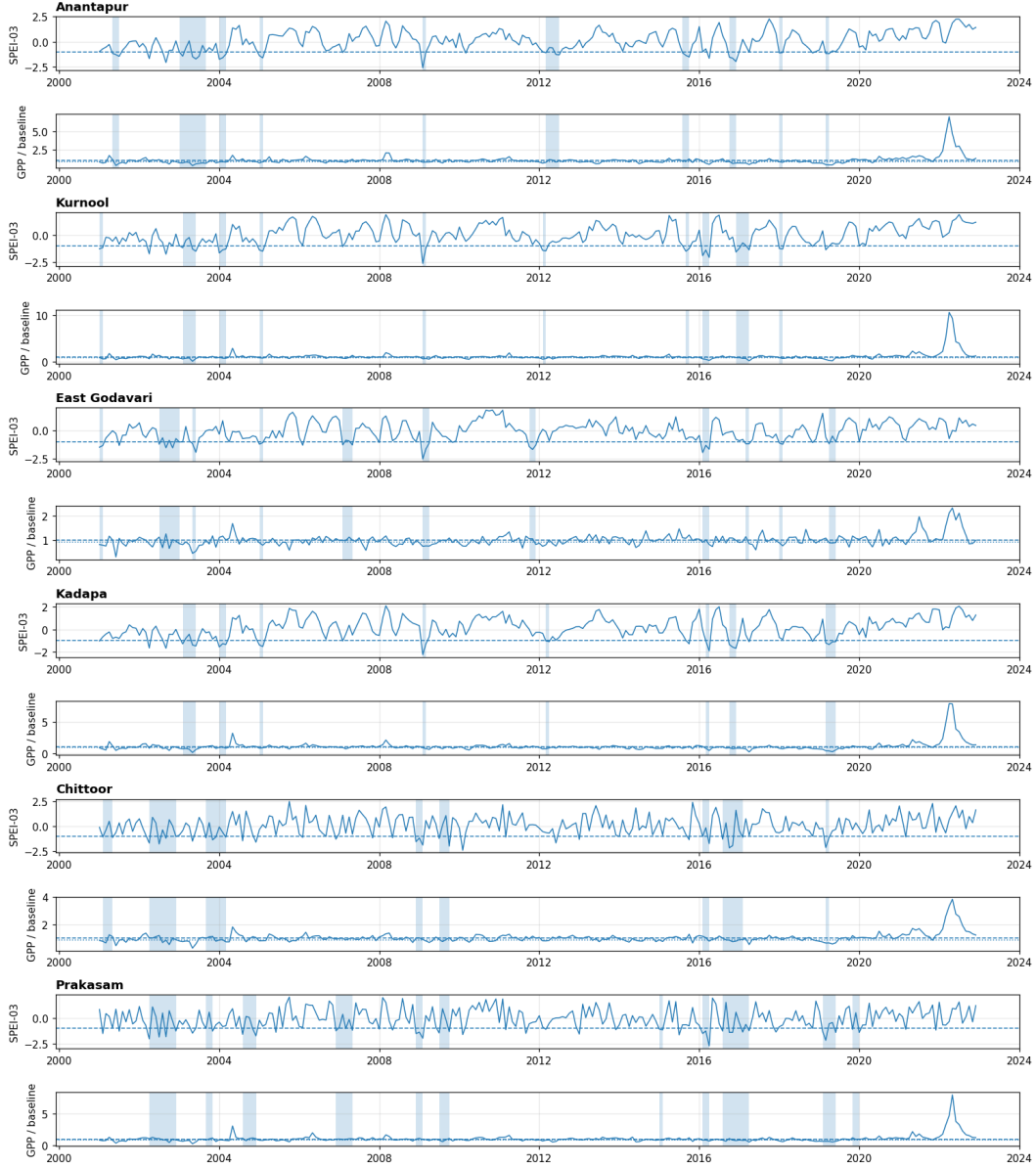


Figure 1: SPEI-03 (top of each pair, dashed line = -1.0 threshold) and GPP relative to seasonal baseline (bottom, dotted line = 0.90) for each district, 2001–2022. Shaded bands are detected drought events.

Figure 1 shows the detected events against the SPEI and GPP records. Droughts cluster in the same years across districts (early 2003–2005, 2009, 2016–2017, and 2019), which is why events in neighbouring districts cannot be treated as fully independent. The panels also show an unusually large GPP spike in 2022 in every district (GPP at several times its baseline); this feature is discussed under limitations.

4.2 Resistance: repeated versus isolated droughts

Outcome	n (rep./iso.)	Mean rep.	Mean iso.	Diff.	Bootstrap 95% CI	MW p	Perm. p
Resistance	2 / 56	0.622	0.651	-0.030	[-0.242, +0.183]	0.982	0.920
Recovery time (months)	2 / 52	3.50	2.85	+0.654	[-2.423, +3.635]	0.751	-
Post-drought residual	2 / 56	-0.117	-0.271	+0.154	[-0.122, +0.430]	0.612	0.583
Recovery gain	2 / 56	0.781	0.318	+0.463	[+0.228, +0.699]	0.068	0.139

Table 4: Repeated (rep.) versus isolated (iso.) droughts. MW = Mann–Whitney U test; Perm. = permutation test stratified by district.

Resistance. Successive droughts did not show a lower resistance than isolated droughts: the mean difference was only -0.030 (0.622 versus 0.651), with a bootstrap confidence interval spanning zero (Table 4, first row). The district-stratified permutation test agrees ($p = 0.920$). In the OLS model adjusting for severity, duration, and district, with year-clustered standard errors, the repeat coefficient was -0.113 (95% CI -0.513 to $+0.287$, $p = 0.581$); drought severity ($p = 0.718$) and duration ($p = 0.109$) were also not significant predictors.

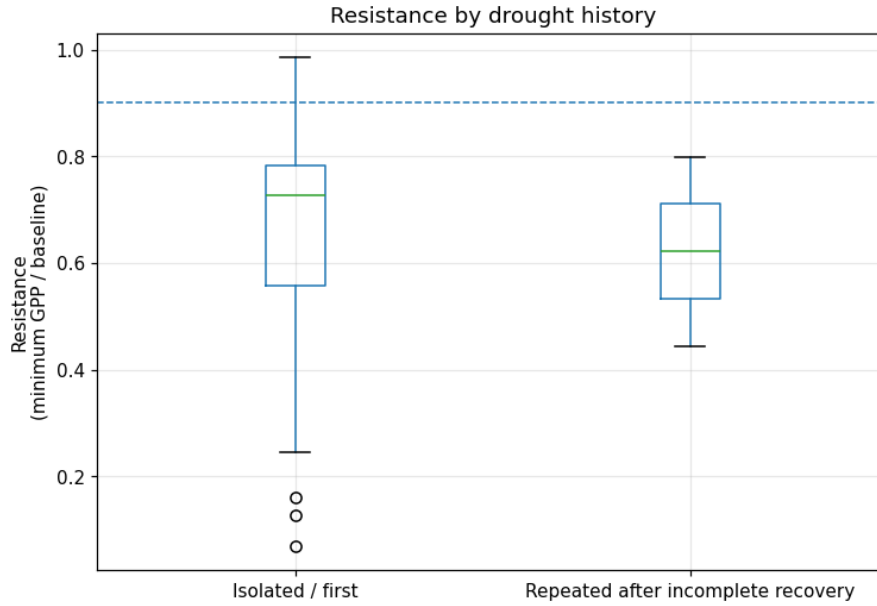


Figure 2: Drought resistance for repeated vs. isolated droughts ($n = 58$: 56 isolated, 2 repeat). Dashed line = 0.90 loss threshold.

Figure 2 compares resistance scores between repeated and isolated drought events. The two repeat events fall within the range of the isolated events, consistent with the absence of a detectable effect in Table 4. Note that with only two repeat events (resistance 0.44 and 0.80) the repeat box simply spans those two values and is not a meaningful distribution. The isolated group includes three very low outliers (resistance below 0.20).

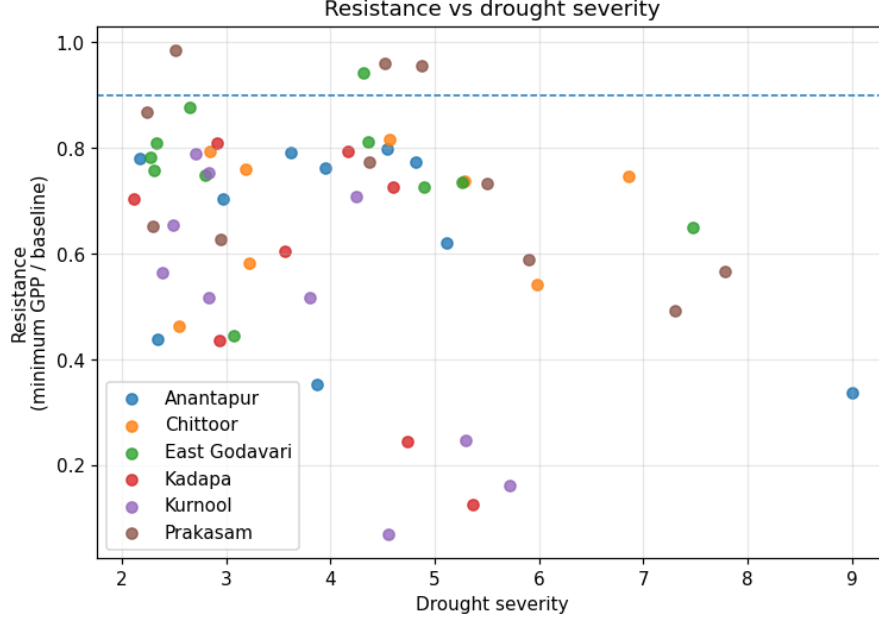


Figure 3: Resistance versus drought severity, coloured by district.

Figure 3 shows resistance against drought severity. There is no clear relationship: events of similar severity span the whole range of resistance, and the lowest values occur at moderate severity (about 4–5) in Kurnool and Kadapa, consistent with the non-significant severity coefficient in the regression.

4.3 Recovery: repeated versus isolated droughts

Recovery time could be computed for 54 events (52 isolated, 2 repeat). The two repeat events took on average 3.5 months to recover versus 2.85 months for isolated events (medians 3.5 and 2.0), a difference of +0.65 months whose confidence interval (-2.42 to $+3.64$) includes zero ($p = 0.751$). The direction is consistent with slower recovery after a successive drought, but with only two repeat events it cannot be distinguished from the natural spread among isolated events.

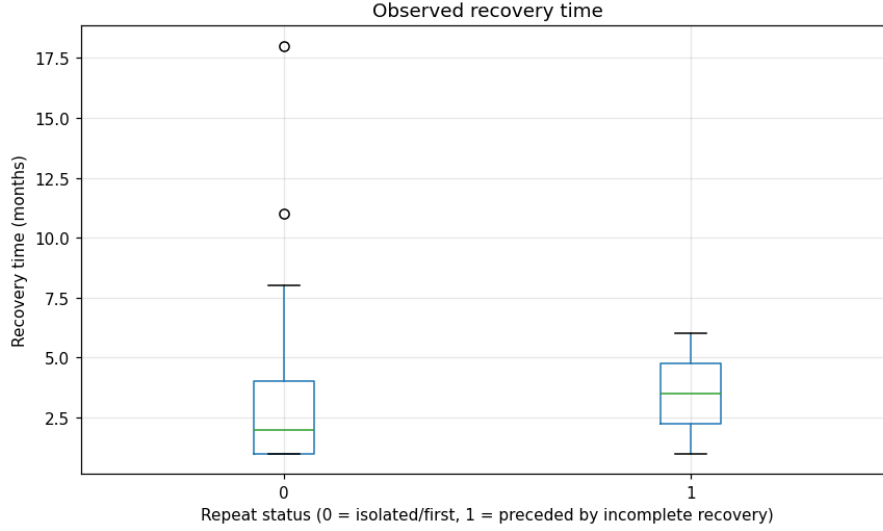


Figure 4: Recovery time for repeated (1) vs. isolated (0) droughts (events with a GPP loss; $n = 54$).

Figure 4 shows recovery time (months to return to 90% of baseline GPP) for the two groups. Recovery gain – how far GPP rebounds after the drought – was larger for the repeat events (0.781 versus 0.318), which is in the opposite direction to what the hypothesis predicts. Its bootstrap interval excludes zero, but the bootstrap is unreliable with only two events in one group, and the Mann–Whitney ($p = 0.068$) and district-stratified permutation ($p = 0.139$) tests do not reach significance. We therefore treat this result as inconclusive rather than as evidence that repeated drought boosts recovery. The earlier draft reported that successive droughts recovered marginally faster; under the revised definitions that pattern is no longer seen for recovery time, which illustrates how sensitive such small-group comparisons are.

4.4 Regional (district-level) comparison

Examining resistance scores grouped by district shows a regional pattern that is weaker and less tidy than in the three-district draft:

District	Events	With loss	Median re- sistance	Mean resis- tance	Median recovery (months)
East Godavari	11	10	0.758	0.753	2.0
Chittoor	8	8	0.743	0.680	2.5
Anantapur	10	10	0.734	0.636	1.5
Prakasam	11	8	0.732	0.746	1.5
Kadapa	8	8	0.654	0.556	1.0
Kurnool	10	10	0.541	0.499	1.0

Table 5: Resistance and recovery by district, ordered by median resistance, summarised from the event table (recovery medians use events with a GPP loss).

- **East Godavari** (coastal, wetter) has the highest median and mean resistance (0.758 and 0.753), but Chittoor, Anantapur, and Prakasam have almost identical medians (0.73–0.74).

- **Kurnool** has the lowest resistance (median 0.541), followed by **Kadapa** (0.654). Relative to Anantapur, Kurnool’s coefficient in the year-clustered regression was -0.177 ($p = 0.065$), Kadapa’s -0.116 ($p = 0.154$), while Prakasam’s was $+0.153$ ($p = 0.030$).

Kurnool has the lowest and East Godavari the highest median resistance, but the highly drought-prone Anantapur is not the lowest, so the data do not reproduce a simple “wetter = more resistant” ordering. The changes in definition (a seasonal, non-drought-month baseline instead of the overall long-term mean, and a minimum taken over the response window) also change absolute resistance values, so the numbers here should not be compared directly with the earlier draft. In all districts the median recovery time is between 1 and 2.5 months; the spread is driven by a few slow events, notably 18 months in Kurnool and 11 months in East Godavari (the 2002–03 drought), and we found no clear evidence that any district recovers systematically more slowly than the others.

4.5 Legacy checks and robustness

Event order and prior-drought effects. If drought damage accumulated over time, later events (or events following severe ones) should show lower resistance or weaker post-drought GPP. Table 6 shows the opposite or no relationship in most cases.

Predictor	Outcome	n	Spearman ρ	p
Event order	Resistance	58	+0.180	0.177
Event order	Recovery time	54	−0.093	0.504
Event order	Post-drought residual	58	+0.343	0.008
Prior-drought severity	Resistance	52	+0.076	0.592
Prior-drought severity	Recovery time	48	+0.036	0.810
Prior-drought severity	Recovery gain	52	+0.289	0.038
Prior-drought severity	Post-drought residual	52	+0.383	0.005
Pre-drought GPP state	Resistance	56	+0.325	0.015
Pre-drought GPP state	Recovery time	52	−0.194	0.168

Table 6: Selected legacy and dose-response checks (Spearman). Twelve tests were run in total; the table lists the main ones.

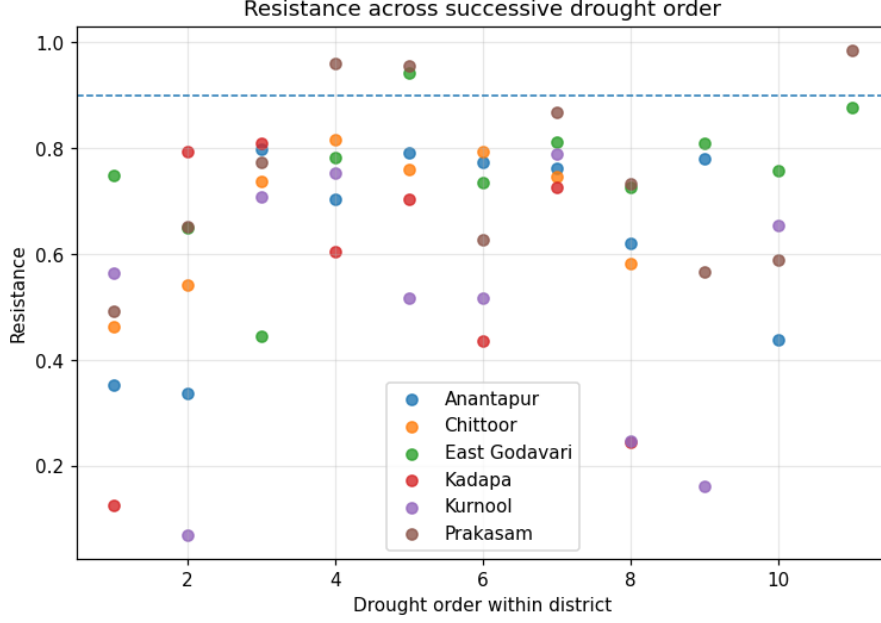


Figure 5: Resistance across successive drought order within each district.

Figure 5 plots resistance against the order of each event within its district. There is no downward trend with order; if anything the later events lie slightly higher, matching the positive (non-significant) correlation in Table 6.

None of the tested predictors was significantly associated with *lower* resistance or *longer* recovery. Several associations went the other way: post-drought GPP was closer to normal for later events and for events following more severe droughts. These should be interpreted cautiously: 12 correlations were examined, so one or two nominally significant results are expected by chance, and event order is confounded with calendar time (we have not yet controlled for a long-term trend in GPP). One nominally significant association was observed between pre-drought GPP state and resistance: ecosystems with higher pre-drought GPP z-scores tended to show higher resistance ($\rho = 0.325$, $p = 0.015$). This association should be interpreted cautiously given the number of exploratory correlations tested.

Cross-verification. GPP-based and NDVI-based resistance agreed moderately across the 58 events (Spearman $\rho = 0.514$, $p < 0.001$): the two indicators rank events similarly but not identically, as expected since GPP and NDVI capture different aspects of vegetation. A Kruskal–Wallis test found no significant difference in resistance by season of onset ($H = 4.76$, $p = 0.190$), indicating that the seasonal baseline removed the seasonal artefact that would otherwise bias events starting in low-GPP months.

5 Observations

5.1 No statistically detectable effect of repeated drought was found, but the hypothesis could be tested only weakly

Across resistance, recovery time, and post-drought response, successive droughts were not detectably worse than isolated ones: the resistance difference was -0.030 (stratified permutation $p = 0.92$), the recovery-time difference $+0.65$ months (CI -2.4 to $+3.6$), and the year-clustered regression gave a repeat coefficient of -0.113 ($p = 0.58$). However, only **two** of the 58 events

qualified as successive under our definition. With so few events the confidence intervals are wide, and the data cannot rule out a moderate legacy effect. The honest conclusion is therefore: **in our six study districts we found no detectable evidence that repeated drought weakens ecosystem resistance or slows recovery.** One possible explanation is that ecosystems often recover relatively quickly (median 2 months), so in many cases the next drought arrives only after recovery has occurred. The study does not have enough repeated events (2 of 58, or 3.4%) to strongly test the research question. The absence of a statistically detectable repeat effect should therefore be interpreted as insufficient evidence to establish an effect, rather than evidence that repeated drought has no effect.

5.2 What the data do show

- **Most events were associated with a GPP loss, but all recovered within the study window.** 93% of events had a GPP loss (resistance below 0.90), and all 54 of these recovered within 24 months, with a median recovery time of 2 months.
- **Repeat exposure before recovery was uncommon in this dataset.** Only 2 of 58 droughts (3.4%) began before recovery from a previous drought. This is a feature of these six districts and this period under our definitions, and should not be generalised.
- **Pre-drought condition may matter.** Ecosystems in better condition before a drought (higher pre-drought GPP z-score) resisted it better ($\rho = +0.33$, $p = 0.015$). This is an association, not a demonstrated cause.

These observations are exploratory. They suggest that resistance and recovery may depend on different factors, and that a repeated-drought framing alone may not capture ecosystem vulnerability, but the present data are too limited to establish this.

5.3 Relationship to our literature review

This finding partially diverges from Xu et al. (2021), who found progressively decreasing ANPP resistance across four consecutive experimentally-imposed, consecutive-year droughts in a semi-arid grassland. A plausible explanation for this divergence is the difference between controlled, back-to-back experimental drought treatments and naturally-occurring, variably-spaced droughts observed via satellite over a 22-year record – real droughts do not recur on a fixed annual schedule, and the time available for recovery between them varies considerably. Our result is more in line with Liu et al. (2026), who found no consistent global legacy effect on GPP, and Gliesch et al. (2025), who found that even substantial, measurable legacy effects (in soil microbial communities) did not propagate meaningfully to ecosystem-scale carbon flux outcomes. This is consistent with the view, raised in our literature review’s critical discussion, that legacy effects may be ecosystem-type and methodology dependent rather than a universal rule, although with only two repeat events we cannot test this directly.

5.4 Regional findings

Kurnool (and, to a lesser extent, Kadapa) showed the lowest resistance, while East Godavari, Chittoor, Anantapur, and Prakasam were similar to one another. Regional differences therefore appear to exist, and they look larger than the (very imprecisely estimated) difference between repeated and isolated droughts, which is consistent with our decision to study multiple districts. However, the pattern is not a simple wet-to-dry gradient, so we avoid claiming one.

5.5 Honest limitations

- **Very few successive events.** Only 2 of 58 events were successive, so the central comparison has low statistical power; only very large effects could be detected reliably.
- **Events are not fully independent.** Several droughts occur in the same years across neighbouring districts (for example, early 2004–2005, early 2009, 2016, and 2019). We used year-clustered standard errors and within-district permutations to partly address this, but the effective sample size is smaller than 58.
- **Definition-dependent results.** The count of successive droughts depends on how recovery is defined (90% of baseline for two consecutive months) and on the SPEI threshold. We have not yet run the full sensitivity analysis over these choices.
- **Multiple testing.** Several nominally significant legacy correlations come from a set of 12 tests, and event order is confounded with calendar time.
- **Scope of data.** The analysis period ends in 2022 so that all districts share the same window, although some GPP records extend to 2026. Rainfall and soil moisture were not used, and temperature was computed for each event but not analysed.
- **Unexplained GPP spike in 2022.** In every district GPP rises to several times its baseline in mid-2022 (Figure 1). This may be a real vegetation response after the wet period, but it may also be a data artefact (for example from stitching two MODIS product versions). Because the z-score anomalies use the mean and standard deviation of the series, such a spike can inflate the variability used for scaling and weaken the post-drought residual and recovery-gain metrics; the ratio-based resistance and recovery time are less affected. This should be checked before the final submission.
- **Events with no GPP loss** (4 of 58) have no recovery time and are excluded from recovery analyses.

5.6 Planned next steps for final submission

- Run a sensitivity analysis over the SPEI threshold, merge gap, and recovery threshold, and report the alternative 24-month-gap definition of “repeat”, which would classify more events as successive and increase statistical power.
- Extend the analysis to the full record available (to 2025–2026) and control for long-term GPP trend when testing event-order effects.
- Consider pixel-level (sub-district) analysis to substantially increase the number of independent data points beyond 58 district-level events.
- Re-introduce soil moisture and temperature as covariates in a regression framework, with confidence intervals, to examine what governs resistance and recovery time, including whether hotter droughts coincide with larger GPP losses.
- Expand the discussion of compound heat-drought stress as a candidate mechanism once temperature is analysed.

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